

Advay Kadam

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EDUCATION

University of Illinois Urbana-Champaign

Champaign, IL

B.S. Computer Science and Statistics, Minor in Mathematics | GPA: 3.9

May 2027

- **Relevant Courses:** Graduate Distributed Systems, Machine Learning Systems, System Programming, Deep Generative Models, Algorithms & Models of Computation, Computer Architecture, Data Structures in C++
- **Awards:** Top 3% at IMC Prosperity (International Trading Challenge), 2x Dean's List
- **Course Assistant:** Data Structures and Algorithms in C++, Statistical Modeling (Applied Regression)

EXPERIENCE

Amazon

Feb 2026 – Present

Incoming Software Development Engineer Intern

New York City, NY

- Incoming SDE (Java/Go) Intern on Amazon Traffic Engineering team, building scalable, high-performance networking and real-time data processing systems that steer and protect customer internet ingress

Bank of New York Mellon

Jun 2025 – Aug 2025

Software Engineer Intern

Jersey City, NJ

- Built a Python FastAPI microservice to serve a LangGraph/LangChain multi-agent workflow (4 agents + orchestrator) that ingests dataset metadata and returns explainable column-level security labels
- Integrated AWS SQS into the FastAPI backend for queueing real-time dataset uploads across LOBs, supporting 500+ dataset requests/day; Optimized chunking and bounded concurrency to cut end-to-end latency by 30%
- Deployed AWS Lambda background workers to consume SQS messages, execute the multi-agent pipeline, and persist 30,000+ dataset column level security labels to an AWS S3 bucket for retrieval

Procter & Gamble

May 2024 – Aug 2024

Software Engineer Intern

Cincinnati, OH

- Optimized a sales ETL pipeline in Azure Databricks, rewriting SQL/PySpark transformations to cut end-to-end runtime by 40% and reduce cluster cost by 25%
- Built an ML data-ingestion pipeline using a fine-tuned YOLOv9 model to classify retail shelf images and generate new structured signals, increasing labeled coverage by 20% and landing outputs in Azure Data Lake

RESEARCH AND CAMPUS INVOLVEMENT

Supercomputing Systems and AI Lab @ UIUC

Aug 2025 – Present

Machine Learning Systems Researcher (PI: Dr. Minjia Zhang)

Urbana, IL

- Developing fused training kernels for Mixture-of-Experts (MoE) models on AWS Trainium using AWS Neuron's Neuron Kernel Interface (NKI), targeting high-frequency transformer primitives to improve training throughput
- Trained and profiled a Mixtral single-layer workload on AWS Trainium using PyTorch Profiler, generating per-op call-frequency and aggregated runtime breakdowns to identify kernel-fusion optimization candidates

National Center for Supercomputing Applications @ UIUC

Aug 2023 – May 2024

Supercomputing Systems Researcher (PI: Dr. Phuong Cao)

Urbana, IL

- Analyzed 2+ PB of real-time network telemetry (Zeek logs) using Python (Pandas/NumPy) and shell pipelines to quantify post-quantum cryptography (PQC) readiness across enterprise-scale SSH traffic
- Built a Flask-based monitoring dashboard that parses Zeek logs and streams live PQC metrics (algorithm distribution, vulnerable sessions, trend breakdowns), enabling rapid visibility into cryptographic risk

PROJECTS

Hybrid Distributed File System (HyDFS) | C, Sockets (UDP/TCP)

- Built Cassandra-inspired distributed file system with consistent hashing and replication across 10 Linux machines
- Integrated a UDP-based gossip heartbeating protocol for decentralized node liveness detection and fault tolerance

Diffusion Model Framework from Scratch | Python, PyTorch, NumPy, Matplotlib, Seaborn, Deep Learning

- Implemented a PyTorch diffusion pipeline with U-Net denoiser, cosine schedule, and DDIM sampling
- Generated MNIST samples from pure noise and evaluated quality; tuned sampling timesteps to speed up inference

TECHNICAL SKILLS

Languages: Python, C, C++, SQL, R, Java

ML Frameworks + Libraries: PyTorch, LangChain/LangGraph, AWS Neuron Kernel Interface, NumPy

Systems + Tools: Linux, Bash, Git, Docker, Networking (TCP/UDP), AWS (SQS, Lambda, EC2), FastAPI, PySpark